Appendix 2. Risk of bias assessment according to the NIH risk of bias tool.

Quality assessment

Observational cohort and cross sectional study	Was the research question or objective in this paper clearly stated?	Was the study population clearly specified and defined?	Was the participation rate of eligible persons at least 50%?	Was a sample size justification, power description, or variance and effect estimates provided?	For the analyses in this paper, were the exposure(s) of interest measured prior to the outcome(s) being measured?	Was the timeframe sufficient so that one could reasonably expect to see an association between exposure and outcome if it existed?	For exposures that can vary in amount or level, did the study examine different levels of the exposure as related to the outcome?	Were the exposure measures (independent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?	Was the exposure(s) assessed more than once over time?	Were the outcome measures (dependent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?	Were the outcome assessors blinded to the exposure status of participants?	Was loss to follow-up after baseline 20% or less?	Were key potential confounding variables measured and adjusted statistically for their impact on the relationship between exposure(s) and outcome(s)?
Bashir et al.[15]	Yes	Yes	CD (cannot determine)	No	NR	Yes	No	No	NR	CD	No	NR (not reported)	No
Jin-young Kim et al.[16]	Yes	Yes	CD	Yes	NR	Yes	No	No	Yes	Yes	No	Yes	Yes
Kang et al.[26] **(2011)**	Yes	Yes	CD	No	NR	Yes	No	No	Yes	Yes	No	Yes	Yes
Perno-Ioanna et al.[30]	Yes	Yes	CD	No	NR	Yes	No	No	NR	Yes	No	Yes	No

Case series studies	Was the research question or objective in this paper clearly stated?	Was the study population clearly specified and defined?	Were the cases consecutive?	Were the subjects comparable?	Was the intervention clearly described?	Were the outcome measures clearly defined, valid, reliable, and implemented consistently across all study participants?	Was the length of follow-up adequate?	Were the statistical methods well-described?	Were the results well-described?
Altay et al.[28]	Yes	Yes	Yes	CD	Yes	Yes	Yes	No	Yes
Apimonbutr et al.[27]	Yes	Yes	CD	CD	NR	Yes	Yes	No	Yes
Bakhach et al.[29]	Yes	Yes	CD	CD	Yes	No	Yes	No	Yes
El Rassi et al.[36]	Yes	Yes	CD	CD	Yes	Yes	Yes	No	Yes
Garcon et al.[32]	Yes	Yes	CD	CD	Yes	Yes	Yes	No	Yes
Hyun Joo Lee et al.[22]	Yes	Yes	CD	CD	Yes	Yes	Yes	Yes	Yes
Karakaplan et al.[25]	Yes	Yes	CD	CD	Yes	Yes	Yes	Yes	Yes
Ta et al.[14]	Yes	Yes	Yes	CD	No	Yes	Yes	Yes	Yes
Littler et al.[33]	Yes	No	CD	CD	Yes	No	Yes	No	Yes
Renson et al.[19]	Yes	No	CD	CD	Yes	Yes	Yes	Yes	Yes
Scheller et al.[31]	Yes	Yes	Yes	CD	Yes	Yes	Yes	No	Yes
Van der wijk et al.[37]	Yes	Yes	CD	CD	Yes	Yes	Yes	Yes	Yes
Yuasa et al.[38]	No	Yes	CD	CD	Yes	No	Yes	No	Yes

Controlled intervention studies	Was the study described as randomized, a randomized trial, a randomized clinical trial, or an RCT?	Was the method of randomization adequate (i.e., use of randomly generated assignment)?	Was the treatment allocation concealed (so that assignments could not be predicted)?	Were study participants and providers blinded to treatment group assignment?	Were the people assessing the outcomes blinded to the participants' group assignments?	Were the groups similar at baseline on important characteristics that could affect outcomes (e.g., demographics, risk factors, co-morbid conditions)?	Was the overall drop-out rate from the study at endpoint 20% or lower of the number allocated to treatment?	Was the differential drop-out rate (between treatment groups) at endpoint 15 percentage points or lower?	Was there high adherence to the intervention protocols for each treatment group?	Were other interventions avoided or similar in the groups (e.g., similar background treatments)?	Were outcomes assessed using valid and reliable measures, implemented consistently across all study participants?	Did the authors report that the sample size was sufficiently large to be able to detect a difference in the main outcome between groups with at least 80% power?	Were outcomes reported or subgroups analyzed prespecified (i.e., identified before analyses were conducted)?	Were all randomized participants analyzed in the group to which they were originally assigned, i.e., did they use an intention-to-treat analysis?
Abrisham et al.[18]	Yes	CD	Yes	No	Yes	Yes	Yes	Yes	Yes	CD	Yes	No	Yes	CD
Acar et al.[35]	No	CD	No	No	No	Yes	Yes	Yes	Yes	CD	Yes	No	No	CD
Gu et al.[24]	No	CD	No	No	No	Yes	Yes	Yes	Yes	CD	Yes	No	Yes	CD
Kang et al.[23] **(2013)**	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	CD
Kumar et al.[17]	Yes	CD	No	No	No	Yes	Yes	Yes	Yes	CD	Yes	No	Yes	No